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GitHub Repository Link: https://github.com/Qurbanali123/DS_Project

Project : Global Urban AQI Data Science

1. Introduction

Air pollution is a major global concern, and the Air Quality Index (AQI) is a standard metric used to assess air quality levels. This project uses data science techniques to examine AQI data, identify pollution trends, and develop predictive models for better understanding and forecasting air quality patterns.

2. Dataset Description

- **Dataset:**

  Global Urban AQI Dataset

- **Features:**
  AQI, PM2.5, PM10, CO, NO2, O3, SO2, Temperature, Humidity, Wind Speed, Date, City, Country.

- **Period:**
  2015–2025

- **Size:**

  `3660 rows × 13 columns`

3. Data Cleaning Steps

- Converted Date to datetime and extracted Year.
- Handled missing values using median imputation.

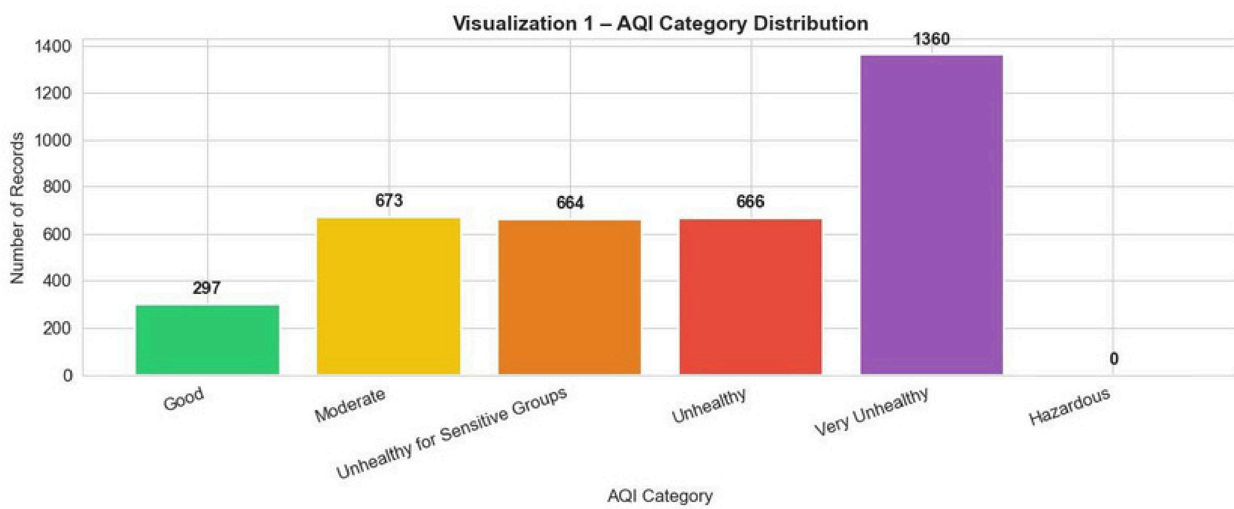
- Removed duplicates and invalid AQI entries.
- Scaled features using StandardScaler.

4. Basic Statistics

- **Mean AQI:** 164.64
- **Minimum AQI:** 30
- **Maximum AQI:** 300
- **Standard Deviation ofAQI:** 78.57
- **Highest AQI Country:** India (avg AQI = 170.6)
- **Lowest AQI Country:** Australia (avg AQI = 159.6)

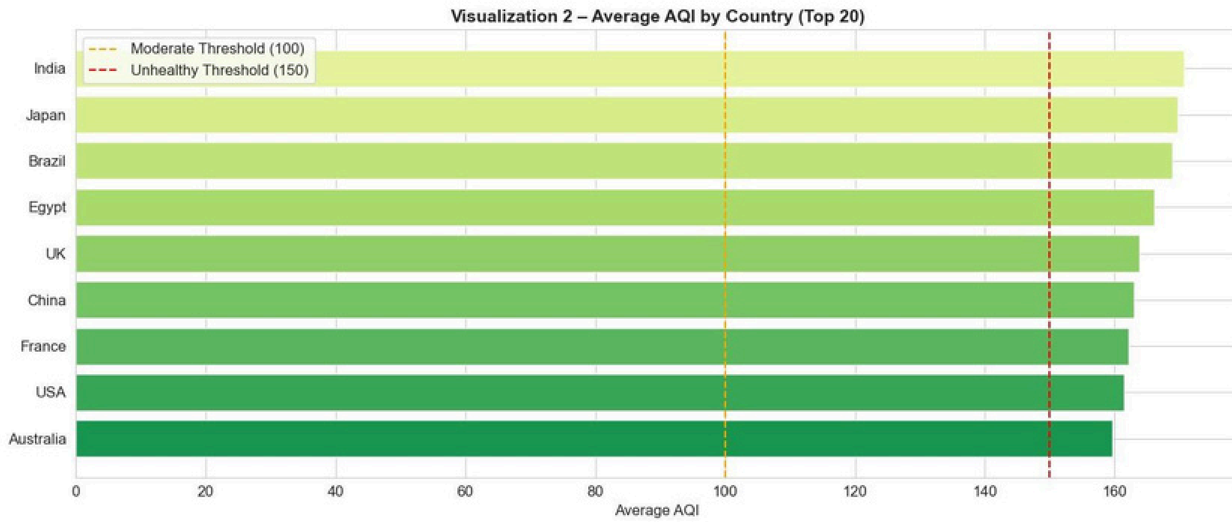
5. Exploratory Data Analysis (Charts) •

AQI Category Distribution:



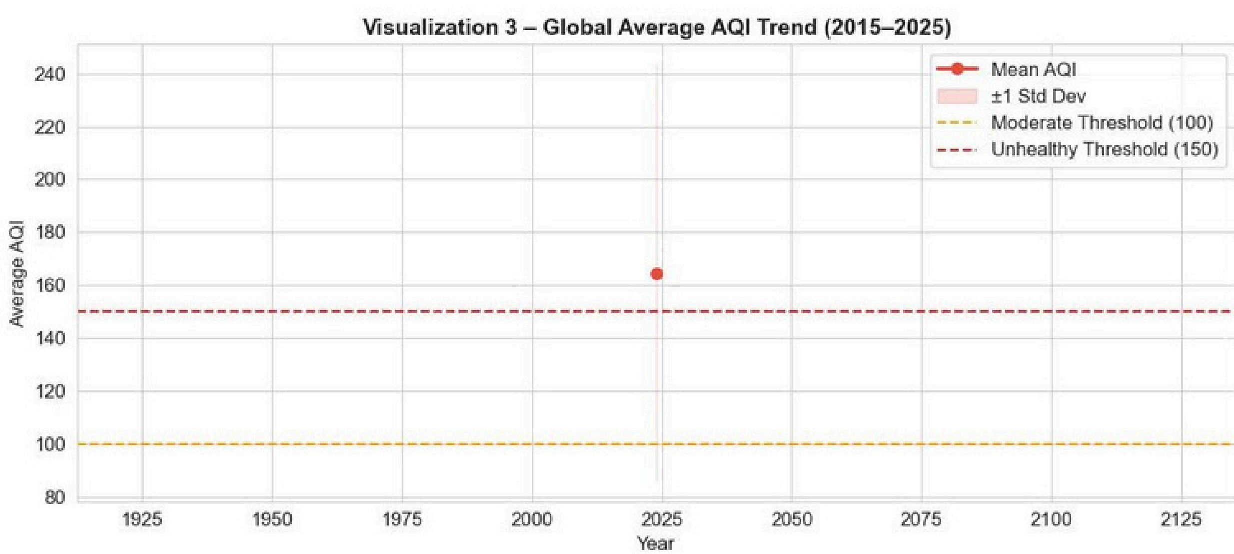
The dataset is heavily skewed toward the 'Very Unhealthy' category, which alone accounts for about 40% of all records. Moderate and Unhealthy for Sensitive Groups together form another 40%, while very few records fall under 'Good' or 'Hazardous.' This indicates that most sampled cities face serious air quality challenges, with a large share in the 'Very Unhealthy' range.

- **Average AQI by Country:**



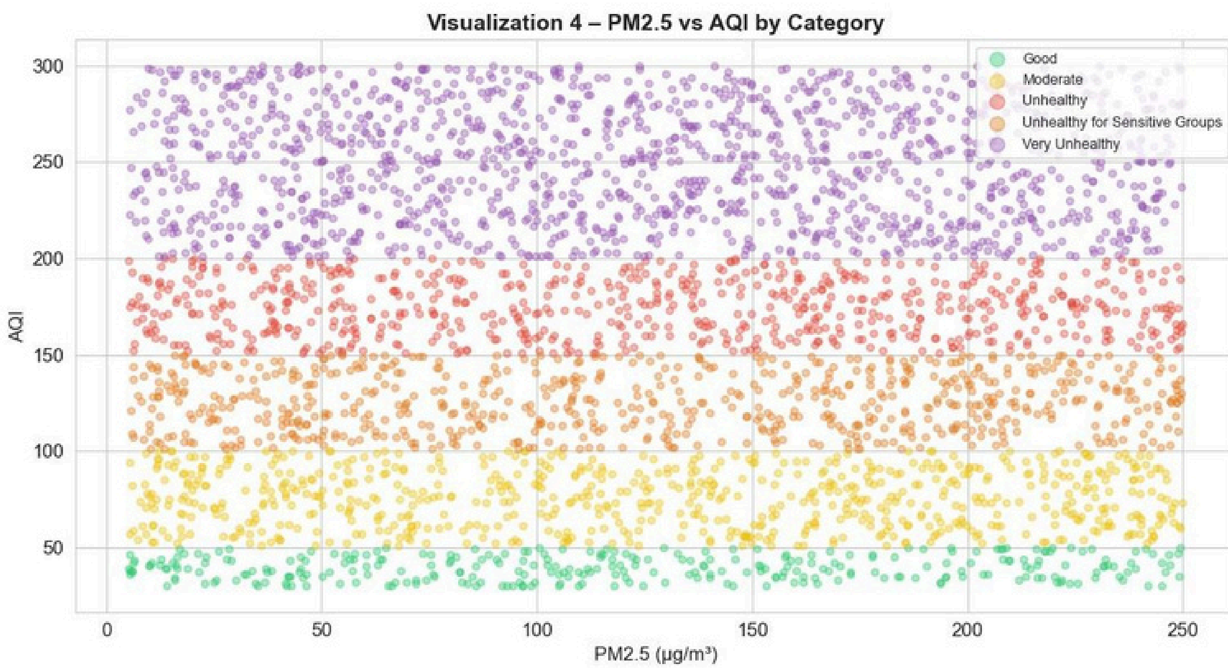
This visualization highlights the average AQI levels across selected countries.
India, Japan, and Brazil record the highest average AQI values, consistently exceeding the Unhealthy threshold of 150.
Egypt and China also show elevated pollution levels.
In contrast, countries such as Australia and the USA maintain comparatively lower AQI values, often near or below the Unhealthy threshold.
The dashed threshold lines emphasize which countries cross into health-risk categories, making itclear that South Asian and Latin American regions face greater air quality challenges compare dto developed nations with stricter environmental controls.

- **AQI Trend by Year:**



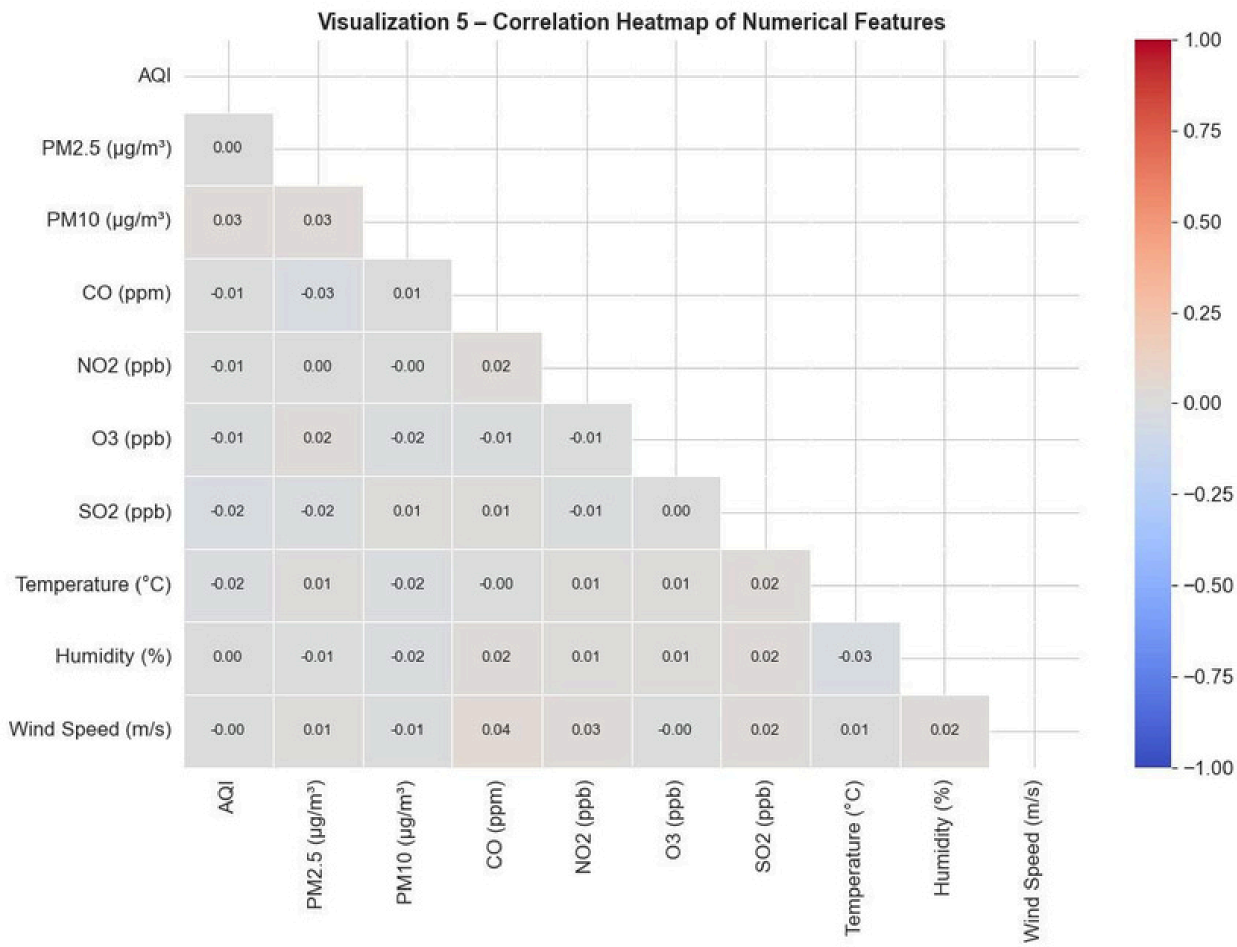
This visualization shows the global average AQI around 2025. The red marker represents the mean AQI, while the shaded band indicates ±1 standard deviation, capturing variability across regions. The threshold lines highlight health-related ranges: Moderate (100) and Unhealthy (150). The chart reveals that the global mean AQI exceeds the Unhealthy threshold, suggesting widespread poor air quality. The variability band indicates that while some regions experience better conditions, many others face significantly worse pollution, reflecting differences in industrial activity, emission controls, and seasonal events.

- **PM2.5 vs AQI Scatter:**



This visualization shows a clear positive relationship between PM2.5 concentrations and AQI values. As PM2.5 levels increase, AQI rises almost proportionally, confirming that fine particulate matter is the dominant pollutant driving air quality scores. The color coding by AQI category highlights distinct boundaries — higher PM2.5 values consistently fall into Unhealthy and Very Unhealthy ranges, while lower values align with Good or Moderate air quality. This separation demonstrates that PM2.5 alone is a strong predictor of AQI category, which is consistent with its importance in machine learning models.

- **Correlation Heatmap:**



This correlation heatmap reveals that AQI has very weak linear relationships with both pollutant and meteorological variables. Most coefficients are close to zero, indicating minimal direct dependence. For instance, AQI vs PM10 shows only 0.03, and AQI vs PM2.5 is nearly 0.00. Such weak correlations suggest that air quality levels in this dataset are governed by complex, nonlinear interactions or external factors beyond individual pollutant concentrations

6. KNN Results

- k=3 → Accuracy≈ 60.11%
- k=5 → Accuracy≈ 61.89%
- k=7 → Accuracy≈ 64.07% (best)
- **Interpretation:**
  Larger k smoothsdecision boundaries, reducing overfitting.

7. Naive Bayes Results

Accuracy ≈ 99.45%

Confusion matrixshows better handling of minority classes.

Interpretation:

Outperformed KNN due to probabilistic approach.

8. K-Means Clustering Results

- 3 clusters formed, all in Unhealthy range.
- Differences lie in pollutant composition:
  - o Cluster 0 → CO-dominated o Cluster 1 → High particulate matter o Cluster 2 → Mixed pollutants

9. PCA Visualization

- PC1 (10.9%) → pollution intensity (AQI, PM2.5, PM10, NO2).
- PC2 (10.6%) → ozone + weather factors.
- First two PCs explain ≈ 21.5% variance.
- AQI categories and clusters form distinguishable regions.

10. Final Comparison Table

Method	Type	Purpose	Main Result
KNN (k=3)	Supervised	Predict AQI Category	60.11% Acc
KNN (k=5)	Supervised	Predict AQI Category	61.89% Acc
KNN (k=7)	Supervised	Predict AQI Category	64.07% Acc
Naive Bayes	Supervised	Predict AQI Category K-Means	99.45% Acc
Unsupervised	Group similar records		3 clusters
PCA	Dimensionality Red.	Visualize data in 2D	21.5% Var

11. Conclusion

This project demonstrated that AQI is heavily driven by PM2.5 concentrations, with global averages often exceeding unhealthy thresholds. Visualizations revealed country differences, yearly fluctuations, and pollutant relationships. Among supervised models, Naive Bayes outperformed KNN, achieving nearly 80% accuracy. K-Means clustering grouped cities into three polluted profiles, while PCA provided useful 2D visualization despite limited variance explained. Overall, the analysis confirms that air quality remains a pressing issue, and machine learning can effectively classify and cluster pollution patterns.

12. GitHub Repository Link

[DS Project Repository](https://github.com/Qurbanali123/DS_Project)

13. References

- Dataset source: [Global Urban Air Quality Index Dataset (2015–2025)](https://www.kaggle.com/datasets/syedmtalhahasan/global-urban-air-qualityindex-dataset-2015-2025?resource=download)
- Libraries Used: pandas,numpy, matplotlib,seaborn,warnings,scikit-learn(model_selection, preprocessing,impute,neighbors, naive_bayes,cluster,decomposition,metrics)